



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Nanded**

Department of Instrumentation Engineering

Summer Term Examination

Academic Year: 2025-26 (Semester – II)

Class: SY B.Tech	Subject: Linear Integrated Circuits	Subject Code: PCC-IN204
Date: 09.06.2026	Time: 14:30 to 17:30	Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/sharing of stationery, calculator, etc. is not allowed
- Non Programmable calculator is allowed.
- Figures to the right indicate full marks.
- The symbols carry the usual meaning.

Q.N.		Attempt All the Questions	Marks	CO	BT
Q.1	A.	What is an integrated circuit. Explain in detail classification of ICs , advantages , disadvantages of ICs and application of ICs.	8	CO1	BT1
	B.	Explain block diagram representation of typical op-amp.. Define different op-amp parametrs.	7	CO1	BT2
Q.2	A.	Determine the output voltage of an differential amplifier for input voltage of $300 \mu V$ and $240 \mu V$. The differential gain of an amplifier is 5000 and value of CMRR is i) 100 ii) 10^5	8	CO1	BT2
	B.	Explain three op-amp Instrumentation amplifier. Also derive output voltage expression for it.	7	CO2	BT1
Q.3	A.	Explain RC phase shift oscillator in detail. Derive relation between output and input voltage.	10	CO3	BT2
	B.	Explain triangular wave generator.	5	CO3	BT2
Q.4	A.	Explain in detail Weinbridge oscillator. Derive expression for resonant frequency Fr .	10	CO3	BT2
	B.	Define filter. Draw the frequency response of any four filters.	5	CO5	BT1
Q.5	A.	Explain in detail first order butterworth low pass filter with its frequency response. Design LPF at cut off frequency 1 KHz with passband gain of 2 . (assume C= 0.01 μf)	8	CO5	BT2 BT5
	B.	Design wide band pass filter with F_L = 200 Hz and F_H = 1 KHz , Pass band gain = 4 . Also find quality factor Q .(Assume C=0.05 μf)	7	CO5	BT6

Q.6	A.	Explain Astable multivibrator using IC555 in detail.	10	CO ₆
	B.	Design astable multivibrator using IC 555 with frequency of 2 KHz and duty cycle of 80 % .	5	CO ₆
Q.7	A.	Explain in detail Phase Locked Loop and Pin diagram of IC 565.	10	CO ₅

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	To understand the operational amplifiers with linear integrated circuits
CO2	To design circuits using operational amplifiers for various applications.
CO3	To infer the DC and AC characteristics of operational amplifiers and its effect on output and their cor techniques
CO4	To Classify and comprehend the working principle of data converters.
CO5	To illustrate the function of application specific ICs such as Voltage regulators, PLL and its app communication.
CO6	To compare the working of multivibrators using special application IC 555 and general purpose Op-amp

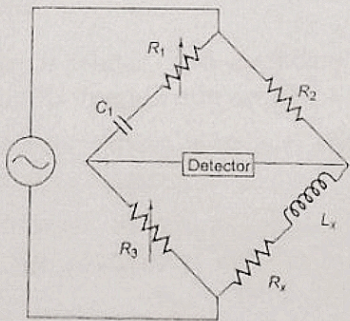


Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Instrumentation Engineering
Summer Term Examination
Academic Year: 2025-26 (semester II)

Class: Second Year	Course: Electrical & Electronics Measurements	Course Code: PCC-IN202
Date: 12.06.2026	Time: 14:30 to 17:30	Max. Marks: 100

Note:

All questions are compulsory.

Q.N.		Marks	CO	BT
Q.1	Define the following terms 1) Sensitivity 2) Accuracy 3) Precision 4) Resolution 5) Linearity 6) Repeatability 7) Reproducibility 8) Range 9) Span 10) Threshold	20	CO1	BT2
Q.2	Explain the errors in measurement and their classification.	10	CO1	BT2
Q.3	Describe the Quality Factor (Q-factor) and Dissipation Factor (D-factor) of inductors and capacitors in detail, with suitable equations and practical examples	10	CO2	BT2
Q.4	Explain the working principle of the Kelvin double bridge with a neat circuit diagram and derive the expression for the unknown resistance. Also mention its advantages, disadvantages, and applications.	10	CO2	BT3
Q.5	Explain how Wien's bridge can be used for measurement of unknown frequency. Draw the phasor diagram under balanced condition and derive the expression for balance.	10	CO2	BT3
Q.6	Describe the constructional details of an electro-dynamometer-type wattmeter. Comment upon the shape of scale when spring control is used.	10	CO3	BT3
Q.7	Explain the working principle of the Maxwell's inductance bridge with a neat circuit diagram. Derive the equation for the unknown inductance and discuss its advantages, disadvantages, and applications.	10	CO2	BT3
Q.8	Find the series equivalent inductance L_x and resistances R_x of the network that causes an opposite angle (hays bridge) to null the following bridge arms. $\omega = 3000 \text{ rad/s}$; $R_2 = 9 \text{ k}\Omega$; $R_1 = 1.8 \text{ k}\Omega$; $C_1 = 0.9 \mu\text{F}$; $R_3 = 0.9 \text{ k}\Omega$. 	10	CO2	BT4
Q.9	Explain the Digital Multimeter (DMM) in detail, including its construction, operation, applications, and a neat, labeled diagram	10	CO4	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO	On successful completion of this course, students should be able to:
CO1	Recalling the basics of measurement, state the purpose of measurement and identify the error sources and minimize its effect in particular measurement.
CO2	Identify and evaluate AC and DC bridges for measurement of R, L, C network.
CO3	Understand the operation, construction and operating principles of electrical instruments.
CO4	Understand the operation, construction and operating principles of electronic instruments.
CO5	Choose appropriate electronic instruments to measure, display and record electrical quantities.



S. G. G. S. Institute of Engineering and Technology, Vishnupuri,
Nanded.

Summer Term Examination , AY 2025-26
Class: SY BTech (Instrumentation Engineering)

Date: 11/06/2026
Time: 14.30 to 17.30 pm

Subject: SENSORS AND TRANSDUCERS
Code: PCC-IN201

Max. Marks 100

Sr. No	Question	Marks	COs	BT
Q. 1	a. Write following statements TRUE or FALSE. I. Strain gauge is a resistive transducer. II. LVDT is passive type of transducer. III. Thermistor is positive resistance temperature coefficient. IV. Piezoelectric Transducer is active transducer. V. Potentiometer Transducer is used for linear displacement measurement.	(05)	CO1, CO3	BT1, BT2
	b. Define following terms: i. Linearity ii. Sensitivity iii. Resolution iv. Accuracy v. Precision	(05)	CO5 CO3	BT1, BT2
Q. 2	a. Design and develop generalized instrumentation system using any sensors for day to day applications.	(05)	CO1, CO3, CO5	BT1, BT2, BT6
	b. For the following given data, $x_1=49.7, x_2=50.1, x_3=50.2, x_4=49.6, x_5=49.7$ calculate the following: 1. Arithmetic mean 2. Deviation 3. Standard deviation	(05)	CO1, CO2	BT3
Q. 3	a. What are the types of errors? Explain each type in detail and how they can be minimised.	(05)	CO1, CO2	BT2, BT1
	b. Draw neat diagram of primary and secondary sensor for pressure measurement. Explain its operation in detail.	(05)	CO2, CO5	BT6, BT3
Q. 4	a. Explain potentiometer transducer for angular displacement measurement with its neat diagram.	(05)	CO1, CO2	BT1, BT2
	b. Explain the construction of strain gauges with the help of neat diagram. What are the parameters can be measured with strain gauge?	(05)	CO1, CO2	BT1, BT2
Q. 5	a. Describe the working of Linear Variable Differential Transformer in detail with its application.	(05)	CO2, CO3	BT1, BT2
	b. Explain capacitive Transducer working principle. Calculate the capacitance of an air capacitor of $3\text{cm} \times 3\text{cm}$ plates, separated by 0.3 mm . $\epsilon=8.9 \times 10^{-12}$.	(05)	CO1, CO2 CO4	BT1, BT5
Q. 6	a. What is Seebeck effect? How EMF is generated in Thermocouple? Explain it with its neat diagram.	(05)	CO1, CO2	BT1, BT2

	b. What is Pt100? A platinum thermometer has a resistance of 110 ohm at 20° C. Find its resistance at 55° C if the platinum has a resistance temperature coefficient of 0.0039/°C.	(05)	CO1, CO2 CO4	BT1, BT5
Q. 7	a. What is manometer? Explain inclined type manometer with its advantages.	(05)	CO1, CO2	BT1, BT2
	b. Draw the schematic diagram of a Bourdon tube and explain its operation.	(05)	CO1, CO2	BT1, BT2
Q. 8	a. Describe with a diagram how flow of liquids can be measured by orifice meter.	(05)	CO1, CO2	BT1, BT2
	b. Explain electromagnetic type flow meter for flow measurement.	(05)	CO1, CO2	BT1, BT2
Q.9	a. Explain level measurement of closed tank by using ultrasonic method.	(05)	CO1, CO2	BT1, BT2
	b. Describe the operation of level measurement with float method.	(05)	CO1, CO2	BT1, BT2
Q.10	Write short notes on following :			
	i. Semiconductor strain gauge	(10)	CO1, CO2	BT1, BT2
	ii. Thermistor		CO1, CO3, CO4	BT1, BT2
	iii. Calibration			
	iv. Bernoulli's equation			
	v. Pyrometer			

Bloom's taxonomy: BT1: Remember BT2: Understand BT3: Apply BT4: Analyse
BT5: Evaluate BT6: Create

Course Outcomes:

1. To identify, list, define verity of sensors, signal conditioning devices, transducer (Primary and secondary).
2. To describe, draw, classify and produced sketches, drawings to explain working principles of various sensors and transducers.
3. To analyze the problem using basic principles for development of project for agriculture, biomedical, Automobiles, Environmental, Petrochemical or other process industries.
4. To monitor and evaluate asses and compare of various sensors and transducers and came to conclusion for the best selection for the desired applications.
5. To create, design, formulate, generate and deliver the solutions for given applications using best applicable sensors and transducers.



Class: SY B.Tech.

Course: Circuit Theory

Course Code: PCCIN203

Date: 08/06/2026

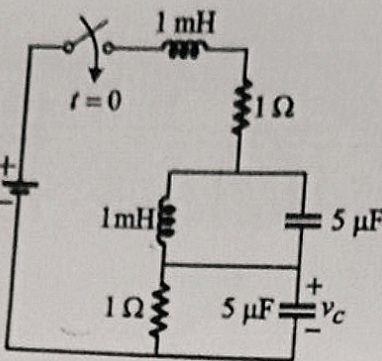
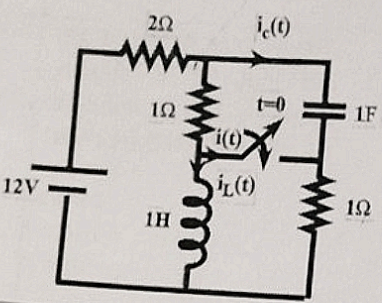
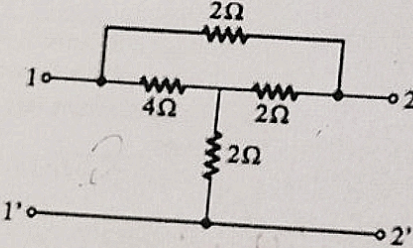
Time: 14:30 to 17:30

Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	The voltage across a 10-mH inductor is shown, sketch the inductor current for $0 < t < 2$ seconds.	10	CO 1	BT2
Q.2	The voltage and current waveforms for an element are shown in figure. The circuit element is _____ and its value is _____.	10	CO 1	BT2
Q.3	Calculate the Value of node voltage V_A in volts and calculate the power absorbed and delivered by each element in the circuit to satisfy the power balance equation $P_{abs} + P_{del} = 0$	20	CO 2	BT3

Q.4	 Calculate the steady state value of voltage v_c if switch is closed at $t=0$ with the help of transient waveforms of voltage and current of passive elements.	20	CO 3	BT3
Q.5	 In the following Figure the switch is opened for a long time and is closed at $t=0$. Determine: $i(0^+)$, $dV_c(t)/dt$ at $t=0^+$ and $di_L(t)/dt$ at $t=0^+$	20	CO 3	BT3
Q6	 For the given two-port network, determine all impedance parameters Z_{11}, Z_{12}, Z_{21}, and Z_{22}, and hence form the impedance matrix $[Z]$. == ALL THE BEST ==	20	CO 4	BT4

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To remember basic concepts and principles of electrical circuits and to solve network problems using mesh current and node voltage equations.
CO 2	To understand and analyse network theorems and their applications
CO 3	To apply and investigate initial conditions and obtain circuit response using Laplace Transform.
CO 4	To evaluate network functions and two port parameters for electrical networks.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Instrumentation Engineering
Summer Term Examination
Academic Year: 2025-26 (Semester – II)

Class: SY	Subject: Sensor Technology	Subject Code: OE-IN202
Date: 20.06.2026	Time: 14:30 to 17:30	Max. Marks: 100

Note:

- i. *ATTEMPT ALL QUESTIONS. Draw neat diagrams wherever necessary.*
- ii. *Assume suitable data whenever necessary and state it clearly.*
- iii. *Mobile phones and programmable calculators are strictly prohibited.*
- iv. *Exchange/sharing of stationery, calculator, etc. is not allowed.*
- v. *Figures to the right indicate full marks.*
- vi. *The symbols carry the usual meaning.*

Q.N.		Marks	CO	BT
Q.1	What is a VOC Gas Sensor? State a few applications of VOC gas sensor.	10	CO1	BT2
Q.2	List the materials commonly used for RTDs. Which one has the most linear characteristics? Draw and explain the construction and working principle of RTD. For a RTD, given that the temperature coefficient of resistance is $0.004/^\circ\text{C}$ at 20°C and $R = 106\ \Omega$ at 20°C . Calculate the resistance at 25°C	10	CO2	BT3
Q.3	Explain the various methods of sensing force? A linear force transducer has an output of 20 mV when measuring 100 N and has an output 120 mV when measuring 300 N. Find the zero offset in mV. (zero offset is the output voltage for zero input force)	10	CO2	BT5
Q.4	In strain gauge based measurements, ideally we would like the resistance of strain gauge to change only in response to the applied strain. However, strain gauge material also responds to changes in the surrounding temperature causing error in the strain measurement. Discuss how the temperature effects can be minimized using Wheatstone bridge.	10	CO3	BT4
Q.5	Compare and contrast between Piezo-electric and piezo-resistive sensors? Explain with neat diagrams.	10	CO1	BT2
Q.6	What are smart sensors? Explain self-calibration and self-testing in smart sensors. What are the advantages and applications of smart sensors?	10	CO5	BT2
Q.7	What is MEMS? What are the components of MEMS? What is so special in Micro-electro-mechanical system (MEMS) that the sensor production technology is revolving around this only? Illustrate with example.	10	CO6	BT2

Q.8		Following are the specifications of a displacement sensor: Measurement range: ± 10 mm, Output range: 4 to 20 mA, Accuracy: $\pm 0.5\%$ FS, Resolution: $3\mu\text{m}$. Calculate: (a) The sensitivity in mA/mm (b) Error in the measurement when subjected to 5mm displacement (c) Smallest change in the output current that can be produced during measurement.	10	CO3	BT1
Q.9		Explain the working principle of LVDT with neat sketch and characteristics. Give advantages, disadvantages and applications of LVDT.	10	CO1	BT2
Q.10		What are the major processing steps in developing the standard semiconductor micro-sensor technology? Describe with suitable diagrams. Enlist the different film deposition processes in this technology and how are they implemented.	10	CO3	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Introduce a variety of sensors, transducers with the state of art technologies.
CO 2	Identify various steps involved in sensor design.
CO 3	Compare to select various sensors for given applications.
CO 4	Design the transducer using appropriate modelling technique.
CO 5	Prepare the testing and calibration documents of sensors.
CO 6	Case study of a MEMS based sensor.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
Academic Year 2025-26 (Semester II)

Summer-Term

Date: 10/06/2026
Time: 14.30-17.30 hrs

Subject : PCC-IN206/PCC-IN207 Signals and Systems
Class : S. Y. (Instrumentation Engg.)

Max. Marks : 100

Note: All Questions are compulsory

Q. No		Question	Marks
Q. 1	A	Given the signal $x[n] = 2 + \cos\left(3\pi n + \frac{\pi}{2}\right) + e^{j\frac{\pi}{3}n}$ Compute the total energy (E_x) and average power (P_x) of the signal. Determine the fundamental period of the signal.	10
	B	Obtain the linear convolution of the following finite length discrete-time signals $x[n]$ and $h[n]$ $x[n] = \{2, 3, 2, 1, 2\}$ and $h[n] = \{1, 3, 1, 1, 2, 2\}$ $\uparrow n = 0$ $\uparrow n = 0$	10
Q. 2	A	Let Input-output relationship of an LTI system is $y[n] = x^2[n]$ Test the system for the following properties. (i) Linearity (ii) Time-Invariance (iii) Causality (iv) Stability and (v) Memoryless	10
	B	Let Impulse response of a continuous-time LTI system is $h(t) = e^{-2t}u(t)$. Obtain the output of the system to unit step input (i.e. when input is $x(t) = u(t)$)	10
Q. 3	A	Determine the Fourier transform of (i) $x(t) = e^{-2t}u(t)$ (ii) $x(t) = \sin(2t)$	10
	B	Consider a causal LTI system with transfer function $H(s) = \frac{1}{(s+2)}$; $\sigma > -2$. Determine the output $y(t)$ of the system when input $x(t) = e^{-4t}u(t)$	10
Q. 4	A	A causal discrete-time LTI system is described by following input-output equation $y[n] = \frac{1}{2}y[n-1] + x[n]$ Determine the output of the system when input is an impulse (i.e. $x[n] = \delta[n]$)	10
	B	A causal continuous-time LTI system is described by following differential equation. $\frac{dy(t)}{dt} + 5y(t) = x(t)$, Find the transfer function and impulse response of the system	10
Q. 5	A	Determine the fundamental frequency ω_0 and Exponential Fourier Series coefficients a_k of the following periodic signal $\tilde{x}(t) = 1 + 3 \cos 2t + 4 \sin 4t$	10
	B	Let the input-output relationship of an LTI system is $y[n] = x[n] + 2x[n-1] + 3x[n-2] + 2x[n-3]$ Determine the impulse response of the system. Also determine the output corresponding to the following input. $x[n] = \{2, 2, 3, 2, 1, 2\}$ $\uparrow n = 0$	10

The End